

Design Verification Plan and Report (virtual test version) — V10a_final_h45

Item	Condition	Result value	Determination	Source
1C discharge capacity	1C, 298.15 K ambient, DFN	5.0618 Ah	PASS (no entry-0 capacity criterion; informational)	r6_final_1c_dfn.json:capacity_ah
-20 C retention	1C discharge, 253.15 K ambient, DFN	0.9944 (vs >= 0.95)	PASS	derived_r6_final_dfn.json
Energy density	contract formula	564.84 Wh/kg (vs >= 327.18)	PASS	r6_final_energy_dfn.json
Volumetric energy density	contract formula	969.50 Wh/L (vs >= 880)	PASS	r6_final_energy_dfn.json
4C fast-charge temperature rise	4C charge, 45 C, lumped thermal, DFN	T_max 329.33 K (vs <= 333.15 K)	PASS	r6_final_4C45_safety_dfn.json:T_max_K
4C fast-charge plating	same, --plating; anode potential < 0 V = plated	anode min +28.96 mV -> plated false	PASS	r6_final_4C45_safety_dfn.json:anode_potential_v
Parameter-set robustness	same design on OKane2022 base (Arrhenius electrolyte)	retention 0.9788; T_max 331.78 K; anode min +24.96 mV	PASS	derived_r6_final_ok_spme.json; r6_final_ok_4C45_safety_dfn.json
Cold-start honesty	Initial temperature 253.15 K (true -20 C equilibrium)	retention 0.9944; T_max 257.10 K	PASS	derived_r6_final_coldstart.json; r6_final_coldstart_lowT_dfn.json
Voltage window	parameter set cut-offs	2.5 – 4.2 V	PASS (set-defined)	Chen2020 set
Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)		
Overcharge to thermal runaway	—	N/A (not simulated)		
Crush / drop	—	N/A (requires physical experiment)		
Cycle life	—	N/A (aging model not run; not simulated)		
Rate-pulse internal resistance	—	N/A (DC resistance 3.367 mOhm reported instead, derived from 1C discharge)	r6_final_energy_dfn.json:dc_cr_ohm	

Conclusion

All simulated verification items PASS against the entry-0 criteria; the design passes on two independent parameter sets at DFN grade. Uncovered items (nail, overcharge-to-thermal-runaway, crush, drop, cycle life) are outside the pure-simulation boundary and require physical experiment.